Ride Pulse- A Bus tracking Software

A SYNOPSIS

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1. Introduction

Public transportation systems in cities often face issues such as lack of real-time information, uncertain bus timings, and difficulty in tracking bus locations. In Guwahati, where a large number of students and working people depend on buses daily, these problems are quite common. Passengers usually wait at bus stops without knowing when the next bus will arrive or whether it has already passed.

Currently, there is no proper centralized system in Guwahati that provides live tracking of buses or accurate timing information. People mostly rely on assumptions or past experience, which is not always reliable.

The proposed Smart Bus Tracking System for Guwahati aims to develop a simple web-based platform where users can track the live location of buses, check routes, and get an estimated arrival time. This system will help improve communication between transport services and passengers, making daily travel more convenient and efficient.

2. Motivation and Objectives

Motivation

- Lack of real-time information about bus locations in Guwahati
- Long waiting time and uncertainty at bus stops
- No centralized platform for tracking public buses
- Need for a simple digital solution for daily commuters

Objectives

- To design and develop a bus tracking system for Guwahati
- To show live location of buses on a map
- To provide estimated arrival time of buses
- To display routes and stops clearly for users
- To create a simple and user-friendly platform

3. Methodology

The project will be developed in the following stages:

Requirement Analysis

- Study the current problems faced by bus users
- Identify important features needed in the system

System Design

- Plan how the system will work overall

- Design layout of the user interface
- Decide how data like routes and locations will be stored

Development

- Create the frontend using basic web technologies
- Develop backend to handle data and processing
- Connect system with map service for location display

Tracking and Prediction Module

- Show live bus location using GPS (or simulated data)
- Estimate arrival time based on distance and movement
- Display routes and stops for user selection

Testing and Implementation

- Test the system to check if all features are working properly
- Make improvements based on feedback
- Demonstrate the working of the system

4. Tangible Outcomes

- A working prototype of bus tracking system for Guwahati
- Live display of bus locations on map
- Estimated arrival time feature
- Clear visualization of routes and stops
- Project documentation including design and testing

5. References for Software Engineering, DBMS, Web Systems

1. Pressman, Roger S. — Software Engineering
2. Silberschatz, Korth, Sudarshan — Database System Concepts
3. Deitel & Deitel — Internet & World Wide Web: How to Program
4. Duckett, Jon — HTML and CSS: Design and Build Websites